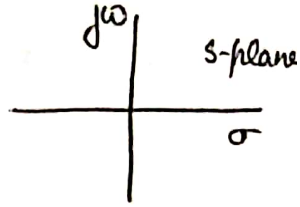
Module - 2

Laplace Transform :-
- transforms time domain diff. eq. into freq. domain algebraic eq
- used to solve differential equations.

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt.$$



$s = \sigma + j\omega$ σ – real
jω – imaginary

$$X(s) = \frac{N(s)}{D(s)} = \frac{b_M s^M + b_{M-1} s^{M-1} + \dots + b_1 s + b_0}{s^N + a_{N-1} s^{N-1} + \dots + a_1 s + a_0}$$

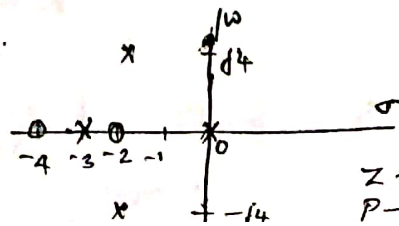
$$= \frac{K(s-s_1)(s-s_2)(s-s_3)\dots(s-s_M)}{(s-s_a)(s-s_b)(s-s_c)\dots(s-s_N)}$$

Roots of the numerator polynomial → Zeroes (o)

Roots of the denominator polynomial → Poles (x)

Poles or zeroes are either real or complex.

complex poles or zeroes always appear in conjugate pairs.



$$\frac{(s+2)(s+4)}{s(s+3)(s+5)(s+2-j4)(s+2+j4)}$$

Z → s = -2, -4
P → s = 0, -3, -5, -2+j4, -2-j4

Laplace Transform of:

→ $x(t) = \delta(t)$

$\delta(t) = 1 \quad t=0$

$$X(s) = \int_{-\infty}^{\infty} \delta(t)\, e^{-st}\, dt$$

$$= \int_{-\infty}^{\infty} 1$$

→ $x(t) = u(t)$

$u(t) = 1 \quad t \geq 0$

$$X(s) = \int_{-\infty}^{\infty} u(t)\, e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-st}\, dt$$

$$= \frac{e^{-\infty} - e^{0}}{-s}$$

$$L[u(t)] = \frac{-1}{s}[-1] = \frac{1}{s}$$

→ $x(t) = e^{-at}\, u(t)$

$$X(s) = \int_{-\infty}^{\infty} e^{-at}\, u(t)\, e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-(s+a)t}\, dt$$

$$= \left. \frac{e^{-(s+a)t}}{-(s+a)} \right]_{0}^{\infty}$$

$$= \frac{1}{-(s+a)}[0-1]$$

$$= \frac{1}{s+a}$$

→ $x(t) = -e^{-at}\, u(-t)$

$u(-t) = 1 \quad t<0$

$$X(s) = \int_{-\infty}^{\infty} -e^{-at}\, u(-t)\, e^{-st}\, dt$$

$$= \int_{-\infty}^{0} -e^{-(s+a)t}\, dt$$

$$= \frac{-1}{-(s+a)} \left. e^{-(s+a)t} \right]_{-\infty}^{0}$$

$$= \frac{1}{s+a}[1-0] = \frac{1}{s+a}$$

$\rightarrow x(t) = u(t-2)$

$$X(s) = \int_{-\infty}^{\infty} u(t-2)\, e^{-st}\, dt$$

$u(t-2) = 1 \quad t \geq 2$

$$= \int_{2}^{\infty} e^{-st}\, dt$$

$$= \frac{-1}{s} \left. e^{-st} \right]_2^{\infty}$$

$$= \frac{-1}{s} \left[0 - e^{-2s} \right]$$

$$= \frac{e^{-2s}}{s}$$

Properties of Laplace Transform

1. Linearity $\quad a x(t) + b y(t) \longleftrightarrow a X(s) + b Y(s)$
2. Scaling. $\quad x(at) \longleftrightarrow \frac{1}{a} X\left(\frac{s}{a}\right) ; \; a > 0$
3. Time Shift. $\quad x(t-\tau) \longleftrightarrow e^{-s\tau} X(s)$.
4. s-domain Shift. $\quad e^{s_0 t} x(t) \longleftrightarrow X(s - s_0)$
5. Convolution $\quad x(t) * y(t) \longleftrightarrow X(s)\, Y(s)$
6. Differentiation $\quad -t x(t) \longleftrightarrow \frac{d}{ds} X(s)$

$\rightarrow$ Find the unilateral laplace transform of

$$x(t) = (e^{-3t} u(t)) * (t\, u(t))$$

A: $e^{-3t} u(t) \xleftrightarrow{L} \frac{-1}{s-3}$

$u(t) \xleftrightarrow{L} \frac{1}{s}$

$t\, u(t) \longleftrightarrow \frac{1}{s^2}$

use Convolution property

$$x(t) = (e^{-3t} u(t)) * (t\, u(t))$$

$$\xleftrightarrow{L} \frac{-1}{s-3} \cdot \frac{1}{s^2}$$

$$X(s) = \frac{-1}{s^2 (s-3)}$$

Initial and final value theorem.

$$x(0^+) = \lim_{s \to \infty} sX(s)$$

$$x(\infty) = \lim_{s \to 0} sX(s)$$

Transfer Function

The transfer function of an LTI system was defined as the Laplace Transform of the impulse response.

$$y(t) = h(t) * x(t)$$

$$Y(s) = H(s) \cdot X(s)$$

$$\boxed{H(s) = \frac{Y(s)}{X(s)}}$$

Transfer function is the ratio of Laplace transform of the output signal to the Laplace transform of input signal with initial condns 0.

Relation between transfer function and differential equation

$$\sum_{K=0}^{N} a_K \frac{d^K}{dt^K} y(t) = \sum_{K=0}^{M} \frac{d^K}{dt^K} x(t) \quad \text{—(1)}$$

If $x(t) = e^{st}$ is the input to an LTI system with impulse response $h(t)$.

$$\begin{aligned} y(t) &= H\{x(t)\} \\ &= h(t) * x(t) \\ &= \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau \end{aligned}$$

As $x(t) = e^{st}$

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} h(\tau)\, e^{s(t-\tau)}\, d\tau \\ &= e^{st} \int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau \end{aligned}$$

$$\boxed{y(t) = e^{st} H(s)}$$

Sub $x(t) = e^{st}$ and $y(t) = e^{st}H(s)$ in ①

$$\sum_{K=0}^{N} a_K \frac{d^K}{dt^k} e^{st} H(s) = \sum_{K=0}^{M} b_k \frac{d^k}{dt^k} e^{st}$$

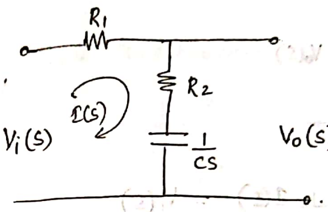
$$\frac{d^k}{dt^k} e^{st} = s^k e^{st}$$

$$\sum_{K=0}^{N} a_K s^k e^{st} H(s) = \sum_{K=0}^{M} b_K \frac{d^k}{dt^k} s^k e^{st}.$$

$$\boxed{H(s) = \frac{\sum_{K=0}^{M} b_k \frac{d^k}{dt^k} s^k}{\sum_{K=0}^{N} a_k \frac{d^k}{dt^k} s^k}} \rightarrow \text{rational transfer func}$$

Modelling a control s/m.

Electrical networks



i R.

← v →

$v(t) = iR$.

$V(s) = I(s) R$.

$$R = \frac{V(s)}{I(s)}$$

L

LS

$$v(t) = L\frac{di(t)}{dt}.$$

$$V(s) = SL\, I(s)$$

$$\frac{V(s)}{I(s)} = LS.$$

C

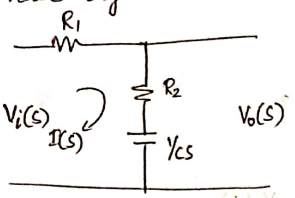
1/cs

$$C\frac{dV(t)}{dt} = i(t) \rightarrow V(t) = \frac{1}{C}\int i(t)dt.$$

$$V(s) = \frac{1}{cs} I(s)$$

$$\frac{V(s)}{I(s)} = \frac{1}{cs}$$

1. Phase lag network.



$$R_1 I(s) + R_2 I(s) + \frac{1}{Cs} I(s) = V_i(s)$$

$$V_o(s) = R_2 I(s) + \frac{1}{Cs} I(s)$$

$$V_i(s) = \left[R_1 + R_2 + \frac{1}{Cs}\right] I(s)$$

$$V_o(s) = \left[R_2 + \frac{1}{Cs}\right] I(s)$$

$$\frac{V_o(s)}{V_i(s)} = \frac{R_2 + 1/Cs}{R_1 + R_2 + 1/Cs}$$

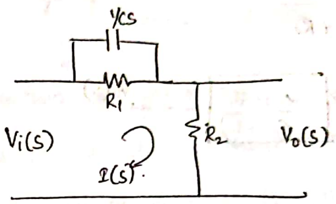
$$= \frac{1/Cs\,[R_2Cs+1]}{1/Cs\,[(R_1+R_2)Cs+1]}$$

$$= \frac{R_2Cs+1}{(R_1+R_2)Cs+1} = \frac{1+sT_1}{1+sT_2}$$

$T_1 = R_2C$
$T_2 = (R_1+R_2)C$

$$V_i(s) \longrightarrow \boxed{\frac{1+sT_1}{1+sT_2}} \longrightarrow V_o(s).$$

2. Phase lead network



$$V_i(s) = \left[\frac{R_1 \frac{1}{Cs}}{R_1 + \frac{1}{Cs}}\right] I(s) + R_2 I(s)$$

$$V_i(s) = \left[\frac{R_1}{R_1Cs+1} + R_2\right] I(s) \quad \text{—— (1)}$$

$$V_o(s) = R_2 I(s) \quad \text{—— (2)}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{R_2}{\frac{R_1 + R_1R_2Cs + R_2}{R_1Cs+1}} = \frac{R_1R_2Cs + R_2}{(R_1+R_2) + R_1R_2Cs}$$

$$= \frac{R_2}{R_1+R_2}\left[\frac{R_1Cs+1}{\frac{R_1R_2}{R_1+R_2}Cs+1}\right] = \alpha\left(\frac{1+sT_1}{1+sT_2}\right)$$

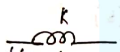
$$\alpha = \frac{R_2}{R_1 + R_2}$$

$$T_1 = R_1 C$$

$$T_2 = \frac{R_1 R_2}{R_1 + R_2}.$$

$$V_i(S) \longrightarrow \boxed{\frac{\alpha(1+ST_1)}{(1+ST_2)}} \longrightarrow V_o(S)$$

Mechanical systems

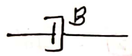


elastic deformation

spring, spring deflection constant, K. restoring force $f_K \propto$ displacement

$f_K = Kx = K(x_1-x_2)$

mass $F_m = ma = M\frac{d^2x}{dt^2}$



damper – dashpot damper.

piston moving inside a cylinder filled with viscous fluid.

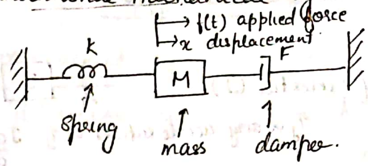
damping force $\propto$ velocity.

coeff. of viscous damping is

$$f_b = B\frac{dx}{dt}.$$

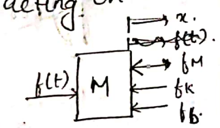
$$\therefore f_b = B\frac{d}{dt}(x_1 - x_2)$$

Translational mechanical system



Application of force $f(t)$ to the mass results in a displacement x (meters). The equation of motion for the system is obtained by Newton's second law of translational motion.

$\therefore$ Inertia force = Forces acting on the mass



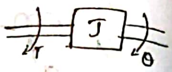
free body dgm

$$f(t) = f_M + f_K + f_b$$

$$= M\frac{d^2x}{dt^2} + Kx + B\frac{dx}{dt}$$

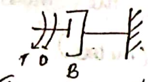
Rotational mechanical system

moment of inertia (J)

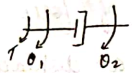


$T_j \propto$ ang. acceleration $\Rightarrow T_j = J\dfrac{d^2\theta}{dt^2}$

Dash pot with rotational frictional coeff. (B)

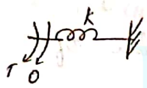


$T_b = B\dfrac{d\theta}{dt}$

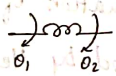


$T_b = B\dfrac{d}{dt}(\theta_1-\theta_2)$

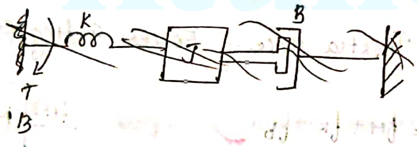
Torsional spring with stiffness (k)

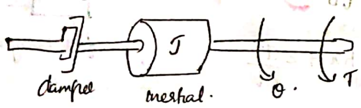


$T_k = K\theta$



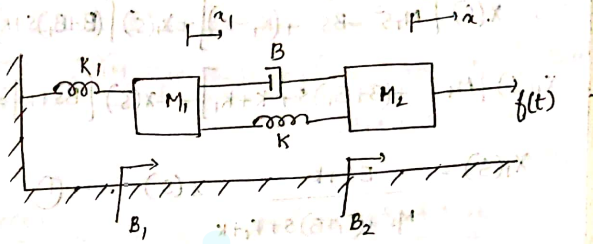
$T_k = K(\theta_1-\theta_2)$





$$T = J\frac{d^2\theta}{dt^2} + B\frac{d\theta}{dt}$$

→ Write the differential equations governing the mechanical systems shown in fig. and determine the transfer function.



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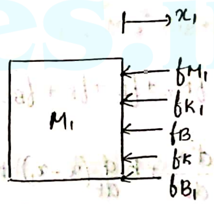


free body dgm of M_1

$$f_{M_1} + f_{K_1} + f_B + f_K + f_{B_1} = 0$$

$$M_1\frac{d^2x_1}{dt^2} + K_1x_1 + B\frac{d}{dt}(x_1-x) + K(x_1-x) + B_1\frac{dx_1}{dt} = 0$$

$$M_1s^2X_1(s) + K_1X_1(s) + Bs(X_1(s) - X(s)) + K(X_1(s) - X(s)) + B_1sX_1(s) = 0$$

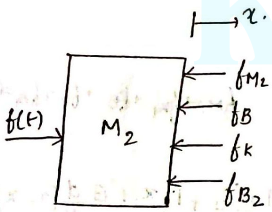
$$X_1(S)\left[M_1S^2 + K_1 + BS + K\right] + B_1 S]$$

$$+ X_0(S)\left[-BS \mp K \cancel{+BS}\right] = 0$$

$$\cancel{X(S)\left[M_1S^2 - BS + (K_1 - K)\right] + X_1(S)\left[(B+B_1)S + K\right] = 0}$$

$$X_1(S)\left[M_1S^2 + (B+B_1)S + K + K_1\right] - X(S)\left[BS + K\right] = 0$$

$$X_1(S) = \frac{BS+K}{M_1S^2 + (B_1+B)S + K_1 + K}\, X(S) \longrightarrow (1)$$



$$f_{M_2} + f_B + f_K + f_{B_2} = f(t)$$

$$M_2\frac{d^2x}{dt^2} + B\frac{d}{dt}(x - x_1) + K(x - x_1) + B_2\frac{dx}{dt} = F(t)$$

$$M_2S^2X(S) + BS\left(X(S) - X_1(S)\right) + K\left(X(S) - X_1(S)\right) + B_2SX(S) =$$

$$X(S)\left[M_2S^2 + BS + K + B_2S\right] + X_1(S)\left[-BS - K\right] =$$

$$X(S)\left[M_2S^2 + BS + K + B_2S\right] - X_1(S)\left[BS + K\right] = F(S)$$

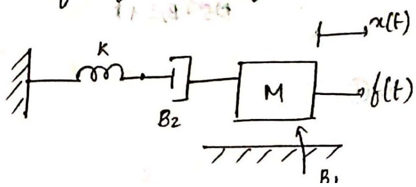
Sub (1)

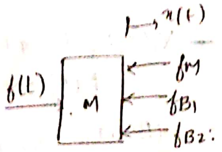
$$X(S)\left[M_2S^2 + (B+B_2)S + K\right] - \frac{(BS+K)^2}{M_1S^2 + (B_1+B)S + K_1 + K}X(S) = F(S)$$

$$\cancel{\frac{F(S)}{X(S)}} \Rightarrow \frac{X(S)\left[M_2S^2 + (B+B_2)S + K\right]\left[M_1S^2 + (B_1+B)S + K_1 + K\right] - (BS+K)^2}{M_1S^2 + (B_1+B)S + K_1 + K} = F(S)$$

$$\frac{X(S)}{F(S)} = \frac{M_1S^2 + (B_1+B)S + K_1 + K}{\left[M_2S^2 + (B+B_2)S + K\right]\left[M_1S^2 + (B_1+B)S + K_1 + K\right] - (B}$$

→ Write the equation of motion in s-domain for the system shown in fig. Determine the transfer function of the system.



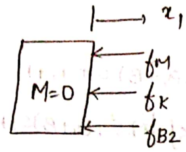


$$f_M + f_{B_1} + f_{B_2} = f(t)$$

$$M\frac{d^2x}{dt^2} + B_1\frac{dx}{dt} + B_2\frac{d}{dt}(x - x_1) = f(t)$$

$$Ms^2X(s) + B_1 s\, X(s) + B_2 s\,[X(s) - X_1(s)] = F(s)$$

$$X(s)\left[Ms^2 + (B_1+B_2)s\right] - B_2 s X_1(s) = F(s) \quad \text{— (1)}$$



$$f_M + f_K + f_{B_2} = 0$$

$$M\frac{d^2x_1}{dt^2} + Kx_1 + B_2\frac{d}{dt}(x_1 - x) = 0$$

$$\cancel{Ms^2}^{\,0} X_1(s) + KX_1(s) + B_2 s X_1(s) - B_2 s X(s) = 0$$

$$X_1(s)\left[\cancel{Ms^2}^{\,0} + K + B_2 s\right] = B_2 s\, X(s)$$

$$X_1(s) = \frac{B_2 s}{\cancel{Ms^2}^{\,0} + B_2 s + K}\, X(s) \quad \text{— (2)}$$

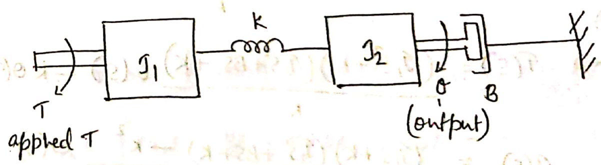
Sub (2) in (1)

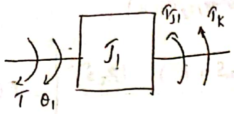
$$X(s)\left[Ms^2 + (B_1+B_2)s\right] - \frac{(B_2 s)^2}{(B_2 s + K)} = F(s)$$

$$X(s)\left[\frac{(Ms^2 + (B_1+B_2)s)(B_2 s + K) - (B_2 s)^2}{(B_2 s + K)}\right] = F(s)$$

$$\frac{X(s)}{F(s)} = \frac{B_2 s + K}{\left((Ms^2 + (B_1+B_2)s)(B_2 s + K)\right) - (B_2 s)^2}$$

→ Write the diff. eq. governing the mechanical rotational system shown in fig. Obtain the transfer function.



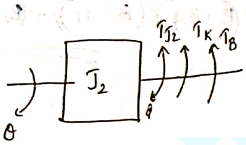


$T = T_{J_1} + T_K$

$T = J_1 \dfrac{d^2\theta_1}{dt^2} + K(\theta_1 - \theta)$

$T(s) = J_1 s^2 \theta_1(s) + K(\theta_1(s) - \theta(s))$

$T(s) = (J_1 s^2 + K)\,\theta_1(s) - K\theta(s)$



$0 = T_{J_2} + T_K + T_B$

$0 = J_2 \dfrac{d^2\theta}{dt^2} + K(\theta - \theta_1) + B\dfrac{d\theta}{dt}$

$0 = J_2 s^2 \theta(s) + K[\theta(s) - \theta_1(s)] + Bs\theta(s)$

$0 = [J_2 s^2 + Bs + K]\,\theta(s) - K\theta_1(s)$

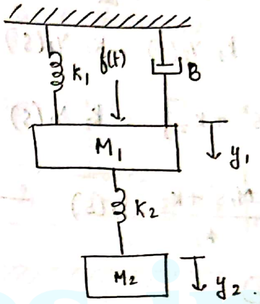
$\theta_1(s) = \dfrac{J_2 s^2 + Bs + K}{K}\,\theta(s)$ — ②

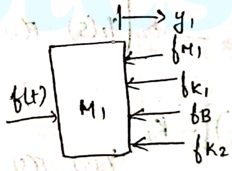
Sub ② ① $\Rightarrow$ $T(s) = \dfrac{(J_1 s^2 + K)(J_2 s^2 + Bs + K)}{K}\,\theta(s) - K\theta(s)$

$T(s) = \dfrac{(J_1 s^2 + K)(J_2 s^2 + Bs + K) - K^2}{K}\,\theta(s)$

$\dfrac{\theta(s)}{T(s)} = \dfrac{K}{(J_1 s^2 + K)(J_2 s^2 + Bs + K) - K^2}$

→ Determine the transfer function $\dfrac{Y_2(s)}{F(s)}$ of the system shown in the fig.



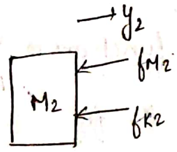


$f_{M_1} + f_{K_1} + f_B + f_{K_2} = f(t)$

$M_1 \dfrac{d^2 y_1}{dt^2} + K_1 y_1 + B\dfrac{dy_1}{dt} + K_2(y_1 - y_2) = f(t)$

$M_1 s^2 Y_1(s) + K_1 Y_1(s) + BsY_1(s) + K_2(Y_1(s) - Y_2(s)) = F(s)$

$Y_1(s)\,[M_1 s^2 + Bs + (K + K_2)]\,Y_1(s) - K_2 Y_2(s) = F(s)$ → ①



$f_{M2} + f_{K2} = 0$

$M_2 \frac{d^2 y_2}{dt^2} + K_2 (y_2 - y_1) = 0$

$M_2 S^2 Y_2(S) + K_2 Y_2(S) - K_2 Y_1(S) = 0$

$Y_2(S) [M_2 S^2 + K_2] = K_2 Y_1(S)$

$Y_1(S) = \frac{M_2 S^2 + K_2}{K_2} Y_2(S)$ —— ②

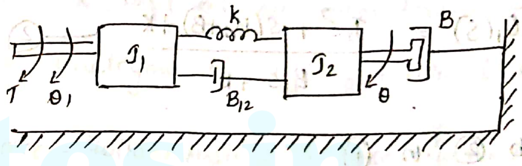
Sub ② in ①

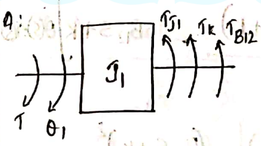
$\left[\frac{(M_2 S^2 + K_2)(M_1 S^2 + BS + (K_1 + K_2))}{K_2} - K_2\right] Y_2(S) = F(S)$

$\frac{[(M_2 S^2 + K_2)(M_1 S^2 + BS + K_1 + K_2)] - K_2^2}{K_2} Y_2(S) = F(S)$

$$\frac{Y_2(S)}{F(S)} = \frac{K_2}{(M_1 S^2 + BS + K_1 + K_2)(M_2 S^2 + K_2) - K_2^2}$$

→ Write the differential equations governing the mechanical rotational system shown in fig. and determine the tranfer functn $\frac{\theta(S)}{T(S)}$



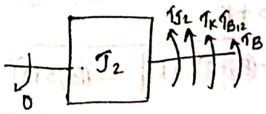


$T_{J_1} + T_K + T_{B12} = T$

$J_1 \frac{d^2\theta_1}{dt^2} + K(\theta_1 - \theta) + B_{12} \frac{d(\theta_1 - \theta)}{dt} = T$

$J_1 S^2 \theta_1(S) + K(\theta_1(S) - \theta(S)) + B_{12} S(\theta_1(S) - \theta(S)) = T(S)$

$\theta_1(S)[J_1 S^2 + B_{12} S + K] - (B_{12} S + K)\theta(S) = T(S)$ —— ①



$T_{J2} + T_K + T_{B12} + T_B = 0$

$$J_2 \frac{d^2\theta}{dt^2} + K(\theta - \theta_1) + B_{12}\frac{d}{dt}(\theta - \theta_1) + B\frac{d\theta}{dt} = 0$$

$$J_2 s^2\theta(s) + K(\theta(s) - \theta_1(s)) + B_{12}s[\theta(s) - \theta_1(s)] + Bs\theta(s) = 0$$

$$\theta(s)\left[J_2 s^2 + B_{12}s + Bs + K\right] - \theta_1(s)\left[K + B_{12}s\right]$$

$$\theta_1(s) = \frac{J_2 s^2 + s(B_{12}+B) + K}{B_{12}s + K} \quad \text{--- (2)}$$

Sub (2) in (1)

$$\left[\frac{(J_1 s^2 + B_{12}s + K)(J_2 s^2 + (B_{12}+B)s + K)}{B_{12}s + K} - B_{12}s + K\right]\theta(s) = T$$

$$\frac{\left[(J_1 s^2 + B_{12}s + K)(J_2 s^2 + (B_{12}+B)s + K)\right] - (B_{12}s + K)^2}{B_{12}s + K}\,\theta(s) = T(s)$$

$$\frac{\theta(s)}{T(s)} = \frac{B_{12}s + K}{\left[(J_1 s^2 + B_{12}s + K)(J_2 s^2 + (B_{12}+B)s + K)\right] - (B_{12}s + K)^2}$$

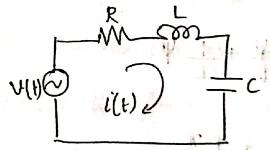
Electrical Analogous of mechanical translational system

It is easier to construct electrical models and analyse them.

M, B, K used in mechanical translational s/m are analogous to R, L, C of electrical s/m.

Force - Voltage Analogy & Force - Current Analogy

F-V analogy



$$v(t) = Ri + L\frac{di}{dt} + \frac{1}{C}\int i\,dt$$

$$i = \frac{dq}{dt}$$

mesh
Series conn.

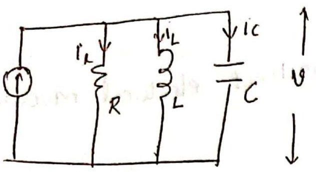
$$V(t) = R\frac{dq}{dt} + L\frac{d^2q}{dt^2} + \frac{1}{C}q$$

$$v(t) = L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q$$

$$f(t) = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + Kx$$

$v \to f$, $i \to q$
$M \to L$, $K \to \frac{1}{C}$
$B \to R$

Force - Current Analogy



nodes parallel.

$$i = i_R + i_L + i_C = \frac{v}{R} + \frac{1}{L}\int v\,dt + C\frac{dv}{dt}$$

$$= \cancel{\frac{dv}{R}} + \cancel{\frac{1}{L}dv}$$

$$v = \frac{d\psi}{dt}$$

$$i = \frac{1}{R}\frac{d\psi}{dt} + \frac{1}{L}\psi + C\frac{d^2\psi}{dt^2}$$

$$i = C\frac{d^2\psi}{dt^2} + \frac{1}{R}\frac{d\psi}{dt} + \frac{1}{L}\psi$$

$$f(t) = M\frac{d^2x}{dt^2} + B\frac{dx}{dt} + kx.$$

$f(t) \rightarrow i$ $\quad M \rightarrow C$

$x \rightarrow \psi$ $\quad B \rightarrow \frac{1}{R}$

$k \rightarrow \frac{1}{L}$

Rotational mechanical system

Torque - voltage analogy.

$T \rightarrow v.$ $\quad M \rightarrow L$

$\theta \rightarrow q.$ $\quad B \rightarrow R$

$k \rightarrow 1/C.$

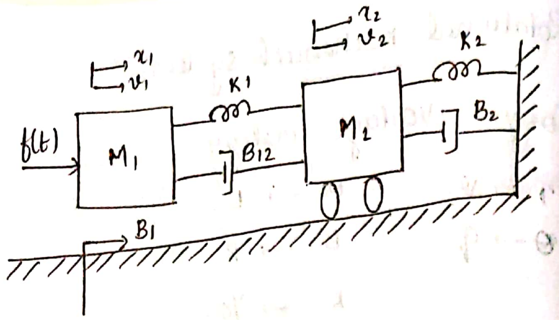
Torque - current analogy

$T \rightarrow i$ $\quad M \rightarrow C$

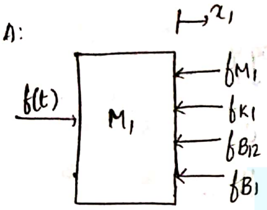
$\theta \rightarrow \psi$ $\quad B \rightarrow 1/R$

$k \rightarrow 1/L$

→ Write the differential equations governing the mechanical system shown in fig. Draw the force - voltage and force - current electrical analogous circuits and verify by writing mesh and node equations.

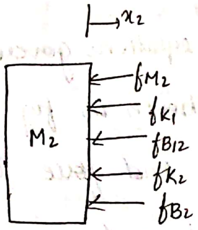


A:



$f_{M_1} + f_{K_1} + f_{B_{12}} + f_{B_1} = f(t)$

$M_1 \frac{d^2x_1}{dt^2} + K_1(x_1-x_2) + B_{12}\frac{d}{dt}(x_1-x_2) + B_1\frac{dx_1}{dt} = f(t)$. —(1).



$f_{M_2} + f_{K_1} + f_{B_{12}} + f_{K_2} + f_{B_2} = 0$

$M_2\frac{d^2x_2}{dt^2} + K_1(x_2-x_1) + B_{12}\frac{d}{dt}(x_2-x_1) + K_2x_2 + B_2\frac{dx_2}{dt}$ —(2)

F-V analogous ckt

2 masses → 2 mesh.

$f(t) \to v(t)$ $M \to L$

$x \to q$ $B \to R$

$v \to i$ $K \to 1/C$.

(1) ⇒ $\cancel{L_1\frac{d^2q}{dt^2}}$ ~~$M_1 \to K_1, B_{12}, 1$~~

~~1st node (M_1) → M_1, B_1~~

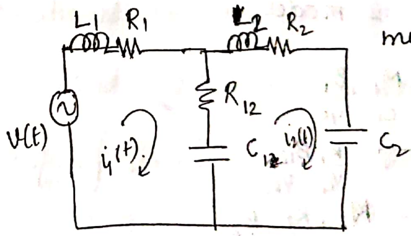
M_1 — K_1, B_{12} — M_2

↓ K_1, B_1 | K_2, B_2.

mesh 1 → M_1, B_1
L_1, R_1

common → K_1, B_{12}
$1/C_1$, R_{12}

mesh 2 → M_2, K_2, B_2
L_2, $1/K_2$, R_2



$$v(t) = L_1\frac{di_1}{dt} + R_1 i_1 + R_{12}(i_1-i_2) + \frac{1}{C}\int(i_1-i_2)dt.$$

$$= L_1\frac{d^2q_1}{dt^2} + R_1\frac{dq_1}{dt} + R_{12}\frac{d}{dt}(q_1-q_2) + \frac{1}{C_1}(q_1-q_2)$$

$$0 = L_2 \frac{di_2}{dt} + R_2 i_2 + \frac{1}{C_2}\int i_2 dt + R_{12}(i_2 - i_1) + \frac{1}{C_1}\int i_2 - i_1 d$$

$$= L_2 \frac{d^2q_2}{dt^2} + R_2 \frac{dq_2}{dt} + \frac{1}{C_2} q_2 + R_{12}\frac{d}{dt}(q_2 - q_1) + \frac{1}{C_1}(q_2 - q_1)$$

Force - Current Analogous circuit

$f(t) \rightarrow i(t)$ $M \rightarrow C$

$x \rightarrow \psi$ $B \rightarrow 1/R$

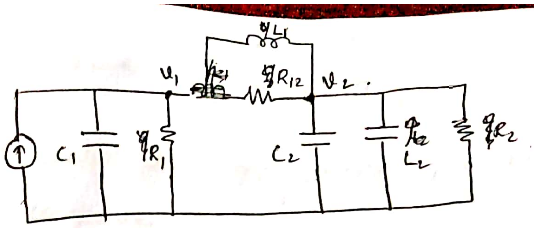
$v \rightarrow v(t)$ $K \rightarrow 1/L$

2 Mass $\rightarrow$ 2 nodes

1st node $\rightarrow$ M_1, B_1
$C_1, 1/R_1$

common $\rightarrow$ K, B_{12}
$1/L_1$ $1/R_{12}$

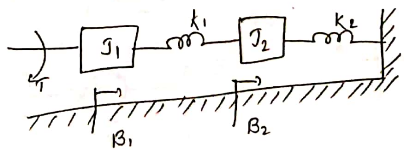
2nd node $\rightarrow$ M_2, K_2, B_2
C_2 $1/L_2$ $1/R_2$



$$C_1 \frac{dv_1}{dt} + \frac{1}{R_1} v_1 + \frac{1}{L_1}\int v_1 - v_2 dt + 1/R_{12}(v_1 - v_2) = i(t)$$

$$\frac{1}{L}\int v_1 - v_2 dt + \frac{1}{R_{12}}(v_1 - v_2) + C_2 \frac{dv_2}{dt} + \frac{1}{L_2}\int v_2 dt + \frac{1}{R_2} v_2 = 0$$

$\rightarrow$ Write the diff. eq. governing the mechanical rotational s/m shown in fig. Draw the T-V and T-I electrical analogous ckts & verify by writing mesh and node eqns.



periodic signals of angular frequencies 0, $\Omega_0 \ldots k\Omega_0 \ldots$. This series of sine and cosine terms is known as trigonometric Fourier series and can be written as

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\Omega_0 t) + b_n \sin(n\Omega_0 t)] \tag{5.4}$$

where a_n and b_n are constants; the coefficient a_0 is called dc component; $a_1 \cos \Omega_0 t + b_1 \sin \Omega_0 t$ the first harmonic, $a_2 \cos 2\Omega_0 t + b_2 \sin 2\Omega_0 t$ the second harmonic and $a_n \cos n\Omega_0 t + b_n \sin n\Omega_0 t$ the n^{th} harmonic.

5.2 Evaluation of Fourier Coefficients

The constants $a_0, a_1, a_2, \ldots a_n, b_1, b_2 \ldots b_n$ are called as Fourier coefficients.

To evaluate a_0 we shall integrate both sides of Eq. (5.4) over one period $(t_0, t_0 + T)$ of $x(t)$ at an arbitrary time t_0.

Thus

$$\int_{t_0}^{t_0+T} x(t)dt = a_0 \int_{t_0}^{t_0+T} dt + a_1 \int_{t_0}^{t_0+T} \left[\sum_{n=1}^{\infty}[a_n \cos(n\Omega_0 t) + b_n \sin(n\Omega_0 t)]dt\right]$$

$$= a_0 T + \sum_{n=1}^{\infty} a_n \int_{t_0}^{t_0+T} \cos(n\Omega_0 t)dt + \sum_{n=1}^{\infty} b_n \int_{t_0}^{t_0+T} \sin(n\Omega_0 t)dt \quad (5.5)$$

Each of the integrals in the summation in Eq. (5.5) is zero since net areas of sinusoids over complete periods are zero for any non zero integer n and any t_0. Thus we obtain

$$\int_{t_0}^{t_0+T} x(t)dt = a_0 T$$

$$\boxed{\begin{aligned} \int_{t_0}^{t_0+T} \cos n\Omega_0 t\, dt = 0 \\ \int_{t_0}^{t_0+T} \sin n\Omega_0 t\, dt = 0 \end{aligned}}$$

(or)

$$a_0 = \frac{1}{T} \int_{t_0}^{t_0+T} x(t)dt \quad (5.6)$$

To evaluate a_n and b_n we use the following results

$$\int_{t_0}^{t_0+T} \cos(n\Omega_0 t)\cos(m\Omega_0 t)dt = \begin{cases} 0 & m \neq n \\ T/2 & m = n \neq 0 \end{cases} \quad (5.7)$$

$$\int_{t_0}^{t_0+T} \sin(n\Omega_0 t)\cos(m\Omega_0 t)dt = 0 \quad \text{for all} \quad m, n \tag{5.8}$$

$$\int_{t_0}^{t_0+T} \sin(n\Omega_0 t)\sin(m\Omega_0 t)dt = \begin{cases} 0 & m \neq n \\ T/2 & m = n \neq 0 \end{cases} \tag{5.9}$$

To find Fourier coefficients a_n, multiply Eq. (5.4) by $\cos(m\Omega_0 t)$ and integrate over one period. That is

$$\int_{t_0}^{t_0+T} x(t)\cos(m\Omega_0 t)dt = a_0 \int_{t_0}^{t_0+T} \cos[(m\Omega_0 t)]dt$$
$$+ \sum_{n=1}^{\infty} a_n \left[\int_{t_0}^{t_0+T} \cos(n\Omega_0 t)\cos(m\Omega_0 t)dt \right]$$
$$+ \sum_{n=1}^{\infty} b_n \left[\int_{t_0}^{t_0+T} \sin(n\Omega_0 t)\cos(m\Omega_0 t)dt \right] \tag{5.10}$$

... integrating

$$\int_{t_0}^{t_0+T} x(t)\cos(m\Omega_0 t)dt = a_m \frac{T}{2} \tag{5.11}$$

Rearranging Eq. (5.11) we arrive at the formula

$$a_m = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(m\Omega_0 t)dt$$

or

$$\boxed{a_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(n\Omega_0 t)dt} \tag{5.12}$$

To find b_n multiply Eq. (5.4) by sin $(m\Omega_0 t)$ we obtain

$$\int_{t_0}^{t_0+T} x(t)\sin(m\Omega_0)dt = \int_{t_0}^{t_0+T} a_0 \sin(m\Omega_0 t)dt$$
$$+\sum_{n=1}^{\infty} a_n \left[\int_{t_0}^{t_0+T} \cos(n\Omega_0 t)\sin(m\Omega_0 t)dt\right]$$
$$+\sum_{n=1}^{\infty} b_n \left[\int_{t_0}^{t_0+T} \sin(n\Omega_0 t)\sin(m\Omega_0 t)dt\right] \quad (5.13)$$

This time the only non zero integral is the last summation in Eq. (5.13), when $m = n$. Thus we have

$$b_m = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(m\Omega_0 t)dt$$

or

$$\boxed{b_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(n\Omega_0 t)dt} \quad (5.14)$$

Example 5.1: Find the trigonometric Fourier series for the periodic signal $x(t)$ in Fig. 5.1.

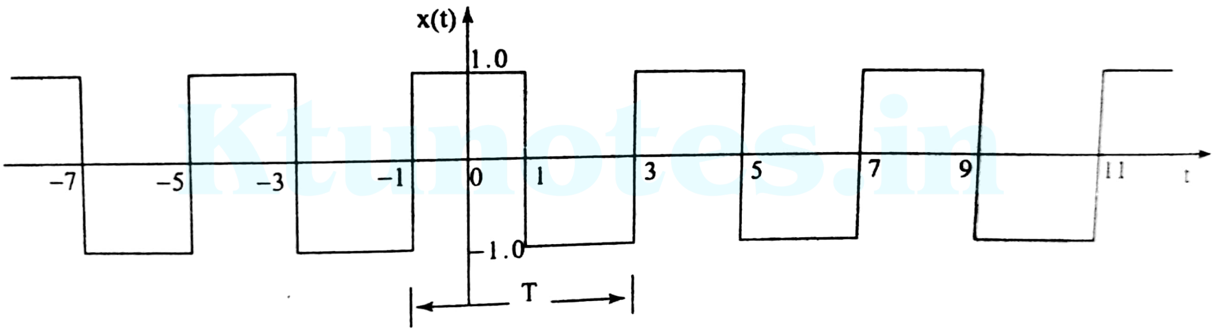


Fig. 5.1

Solution For the given signal the period $T = 4$. For our convenience we choose one period of the signal from $t = -1$ to $t = 3$ rather than from $t = 0$ to $t = 4$ to reduce the number of integrals.

The fundamental frequency

$$\Omega_0 = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$$

and

$$x(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi}{2}t\right) + b_n \sin\left(\frac{n\pi}{2}t\right)\right] \qquad (5.15)$$

Given $x(t) = \begin{cases} 1 & \text{for} \quad -1 \le t \le 1 \\ -1 & \text{for} \quad 1 < t \le 3 \end{cases}$

$$a_0 = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)dt$$

$$= \frac{1}{4}\int_{-1}^{3} x(t)dt = \frac{1}{4}\left[\int_{-1}^{1} dt + \int_{1}^{3}(-dt)\right]$$

$$= \frac{1}{4}\left[t\Big|_{-1}^{1} + (-t)\Big|_{1}^{3}\right] = \frac{1}{4}[1-(-1)-(3-1)] = 0$$

$$\boxed{\because \quad a_0 = 0} \qquad (5.16)$$

$$a_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(n\Omega_0 t)\,dt$$

$$= \frac{1}{2}\int_{-1}^{3} x(t)\cos\left(\frac{n\pi}{2}t\right)dt$$

$\frac{1}{n\pi}\left[\sin\frac{\pi}{2} + \sin\frac{n\pi}{2} - \left\{\sin\frac{3n\pi}{2} - \sin\frac{n\pi}{2}\right\}\right.$

$$= \frac{1}{2}\left[\int_{-1}^{1}\cos\left(\frac{n\pi}{2}t\right)dt + \int_{1}^{3}(-1)\cos\left(\frac{n\pi}{2}t\right)dt\right]$$

$$= \frac{1}{2}\left[\frac{2}{n\pi}\sin\left(\frac{n\pi}{2}t\right)\Big|_{-1}^{1} - \frac{2}{n\pi}\sin\left(\frac{n\pi}{2}t\right)\Big|_{1}^{3}\right]$$

$$= \frac{1}{2}\left[\frac{8}{n\pi}\sin\left(\frac{n\pi}{2}\right)\right] = \frac{4}{n\pi}\sin\left(\frac{n\pi}{2}\right)$$

$$\boxed{a_n = \frac{4}{n\pi}\sin\left(\frac{n\pi}{2}\right)} \tag{5.17}$$

$$b_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(n\Omega_0 t)\,dt = \frac{1}{2}\left[\int_{-1}^{3} x(t)\sin\left(\frac{n\pi}{2}t\right)dt\right]$$

$$= \frac{1}{2}\left[\int_{-1}^{1} \sin\left(\frac{n\pi}{2}t\right) dt + \int_{1}^{3} (-1) \sin\left(\frac{n\pi}{2}t\right) dt\right]$$

$$\frac{1}{2}\left[\frac{-2}{n\pi} \cos\left(\frac{n\pi}{2}t\right)\Bigg|_{-1}^{1} + \frac{2}{n\pi} \cos\left(\frac{n\pi}{2}t\right)\Bigg|_{1}^{3}\right]$$

$$= \frac{1}{2}\left[\frac{-2}{n\pi}\left(\cos\frac{n\pi}{2} - \cos\frac{n\pi}{2}\right) + \frac{2}{n\pi}\left(\cos\frac{3n\pi}{2} - \cos\frac{n\pi}{2}\right)\right]$$

$$= 0 \quad \boxed{b_n = 0} \tag{5.18}$$

$$a_n = \frac{4}{n\pi} \sin\frac{n\pi}{2} = \begin{cases} 0 & n \text{ even} \\ \dfrac{4}{n\pi} & n = 1, 5, 9, 13, \ldots \\ \dfrac{-4}{n\pi} & n = 3, 7, 11, 15, \ldots \end{cases} \tag{5.19}$$

$$
\begin{aligned}
x(t) &= \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{2}\right) t \\
&= \frac{4}{\pi}\cos\left(\frac{\pi}{2}t\right) - \frac{4}{3\pi}\cos\left(\frac{3\pi}{2}t\right) + \frac{4}{5\pi}\cos\left(\frac{5\pi}{2}t\right) - \frac{4}{7\pi}\left(\frac{7\pi}{2}t\right) + \cdots \\
&= \frac{4}{\pi}\left[\cos\left(\frac{\pi}{2}t\right) - \frac{1}{3}\cos\left(\frac{3\pi}{2}t\right) + \frac{1}{5}\cos\left(\frac{5\pi}{2}t\right) - \frac{1}{7}\cos\left(\frac{7\pi}{2}t\right) + \cdots\right. \qquad (5.20)
\end{aligned}
$$

5.3 Symmetry Conditions

From the above two examples we can find some interesting observations. In example (5.1) the Fourier series coefficients $b_n = 0$ and in example (5.2) the Fourier series coefficients $a_0 = a_n = 0$. This is due to the fact that in both examples the waveforms have some type of symmetry.

We already studied that any signal $x(t)$ can be splitted into even and odd functions. That is

$$x(t) = x_e(t) + x_o(t) \tag{5.27}$$

The even and odd parts of the signal $x(t)$ can be obtained from the following relations

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)] \tag{5.28}$$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)] \tag{5.29}$$

These relations can be used to find Fourier coefficients. To have a convenient form, choose the interval of integration from $\frac{-T}{2}$ to $\frac{T}{2}$. Now Eq. (5.12) can be written as

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\cos(n\Omega_0 t)dt$$

and Eq. (5.14) can be written as

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} x(t)\sin(n\Omega_0 t)dt$$

Substituting $x(t) = x_e(t) + x_o(t)$ in the above Eq. (5.30) yields

$$a_n = \frac{2}{T}\left[\int_{-T/2}^{T/2} x_e(t)\cos(n\Omega_0 t)dt + \int_{-T/2}^{T/2} x_o(t)\cos(n\Omega_0 t)dt\right].$$

Similarly

$$b_n = \frac{2}{T}\left[\int_{-T/2}^{T/2} x_e(t)\sin(n\Omega_0 t)dt + \int_{-T/2}^{T/2} x_o(t)\sin(n\Omega_0 t)dt\right]$$

We know that

$$\text{odd function} \times \text{odd function} = \text{even function}$$
$$\text{even function} \times \text{even function} = \text{even function}$$
$$\text{even function} \times \text{odd function} = \text{odd function}$$

and that for any even function $x_e(t)$

$$\int_{-t_0}^{t_0} x_e(t)dt = 2\int_{0}^{t_0} x_e(t)dt \tag{5.34}$$

and for any odd function $x_o(t)$

$$\int_{-t_0}^{t_0} x_o(t)dt = 0 \tag{5.35}$$

If $x(t)$ is an even function, then $x_o(t) = 0$. Substituting this in Eq. (5.32) for b_n yields

$$b_n = \frac{2}{T}\left[\int_{-T/2}^{T/2} x_e(t)\sin(n\Omega_0 t)dt\right] \tag{5.36}$$

Since $x_e(t)$ is even and $\sin(n\Omega_0 t)$ is odd for all n, the product $x_e(t)\sin(n\Omega_0 t)$ becomes odd, resulting integral is equal to zero. Therefore $b_n = 0$. Applying the same condition to Eq. (5.32) for a_0 and a_n results

$$a_n = \frac{4}{T}\int_{0}^{T/2} x(t)\cos(n\Omega_0 t)dt \tag{5.37}$$

and

$$a_0 = \frac{2}{T}\int_{0}^{T/2} x(t)dt \tag{5.38}$$

Thus the Fourier series expansion of an even periodic function contains only cosine terms and a constant.

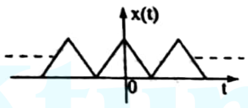
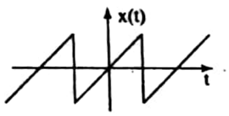
If $x(t)$ is an odd function, then $x_e(t) = 0$. Substituting this in Eq. (5.32) for a_n results

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} x_o(t) \cos(n\Omega_0 t) dt \tag{5.39}$$

Since $\cos(n\Omega_0 t)$ is even the product in the above integral is odd, implying that $a_n = 0$. Similarly we can show that $a_0 = 0$. Applying these conditions to Eq. (5.33) for b_n we obtain

$$b_n = \frac{4}{T}\int_0^{\pi/2} x(t)\sin(n\Omega_0 t)dt \tag{5.40}$$

Table 5.1 Symmetry conditions for Periodic signals

Type of symmetry	Condition	Example	a_0	a_n	b_n	Property
Even	$x(t) = x(-t)$		$a_0 = \frac{2}{T}\int_0^{T/2} x(t)dt$ $\Omega_0 = \frac{2\pi}{T}$ T is fundamental period	$\frac{4}{T}\int_0^{T/2} x(t)\cos(n\Omega_0 t)dt$	0	Cosine terms only
Odd	$x(t) = -x(-t)$		0	0	$\frac{4}{T}\int_0^{T/2} x(t)\sin(n\Omega_0 t)dt$	Sine terms only

5.4 Cosine Representation

The trigonometric Fourier series of Eq. (5.4) contains sine and cosine terms of the same frequency.

Using trigonometric identity

$$a_n \cos(n\Omega_0 t) + b_n \sin(n\Omega_0 t) = A_n\left[\cos(n\Omega_0 t + \theta_n)\right] \tag{5.42}$$

we obtain cosine representation of $x(t)$ which contains sinusoids of frequencies $\Omega_0, 2\Omega_0, \cdots$. That is

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(n\Omega_0 t + \theta_n) \tag{5.43}$$

where

$$A_0 = a_0 \tag{5.44a}$$

$$A_n = \sqrt{a_n^2 + b_n^2} \quad \text{and} \tag{5.44b}$$

$$\theta_n = -\mathrm{Tan}^{-1}\left(\frac{b_n}{a_n}\right) \tag{5.44c}$$

The number A_n are called amplitude coefficients of the Fourier series and the number θ_n are called the phase coefficients the Fourier series.

Example 5.3: Find the cosine representation Fourier series for the signal shown in Fig. 5.3.

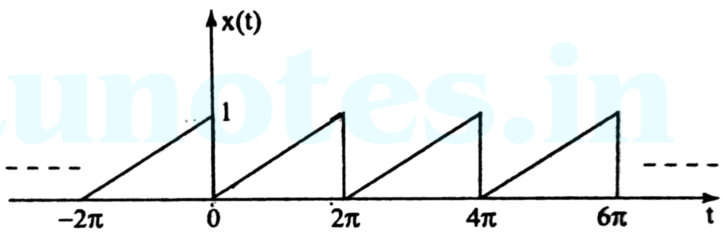


Fig. 5.3

Solution Time period $T = 2\pi$

$$\Omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$$

The signal $x(t)$ over a time period T can be defined as

$$x(t) = \frac{t}{2\pi} \quad \text{for} \quad 0 \le t < T \tag{5.45}$$

$$a_0 = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)dt = \frac{1}{2\pi}\int_0^{2\pi} \frac{t}{2\pi}dt = \frac{1}{4\pi^2}\left.\frac{t^2}{2}\right|_0^{2\pi} = \frac{1}{2} \tag{5.46}$$

$$a_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\cos(n\Omega_0 t)dt$$

$$= \frac{1}{\pi}\int_0^{2\pi}\frac{t}{2\pi}\cos(nt)dt = \frac{1}{2\pi^2}\left[t\left.\frac{\sin nt}{n}\right|_0^{2\pi} + \left.\frac{\cos nt}{n^2}\right|_0^{2\pi}\right] = 0 \tag{5.47}$$

$$b_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin(n\Omega_0 t)dt = \frac{1}{\pi}\int_0^{2\pi}\frac{t}{2\pi}\sin(nt)dt$$

$$= \frac{1}{2\pi^2}\left[-t\left.\frac{\cos nt}{n}\right|_0^{2\pi} + \left.\frac{\sin nt}{n^2}\right|_0^{2\pi}\right]$$

$$= \frac{1}{2\pi^2}\left[-2\pi\frac{\cos 2\pi n}{n}\right] = \frac{-1}{n\pi} \tag{5.48}$$

Using Eq. (5.44)

$$A_0 = a_0 = \frac{1}{2} \tag{5.49}$$

$$A_n = \sqrt{a_n^2 + b_n^2} = \frac{1}{n\pi} \tag{5.50}$$

$$\theta_n = \tan^{-1}\left[\frac{-\left(\frac{-1}{n\pi}\right)}{0}\right] = \frac{\pi}{2} \tag{5.51}$$

Therefore

$$x(t) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{1}{n\pi} \cos\left(nt + \frac{\pi}{2}\right) \tag{5.52}$$

ample 5.4: Find cosine Fourier series of half rectified sine wave

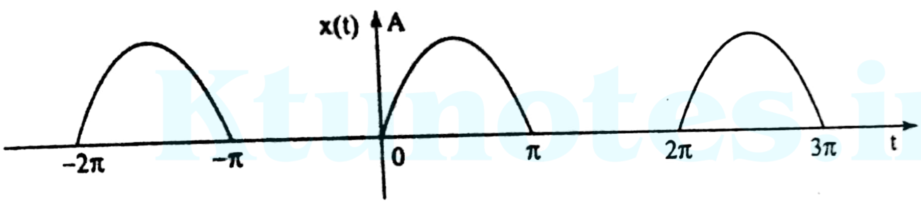


Fig. 5.4

Solution The signal $x(t) = A\sin\Omega_0 t$ for $0 \le t \le \pi$
$= 0$ for $\pi \le t \le 2\pi$

The time period of the wave form is given by $T = 2\pi$ and

$$\Omega_0 = \frac{2\pi}{2\pi} = 1 \tag{5.53}$$

$$a_0 = \frac{1}{T}\int_{t_0}^{t_0+T} x(t)dt = \frac{1}{2\pi}\int_0^{\pi} A\ \sin\ tdt = \frac{-A}{2\pi}\cos\ t\Big|_0^{\pi}$$

$$= \frac{A}{\pi} \tag{5.54}$$

$$a_n = \frac{2}{\pi}\int_{t_0}^{t_0+T} x(t)\ \cos\ ntdt$$

$$= \frac{2A}{2\pi}\left[\int_0^{\pi}\sin\ t\ \cos\ ntdt\right] = \frac{A}{2\pi}\left[\int_0^{\pi}\sin\ (1+n)t + \sin\ (1-n)t\right]dt$$

$$= \frac{A}{2\pi}\left[-\frac{\cos\ (1+n)t}{1+n}\Big|_0^{\pi} - \frac{\cos\ (1-n)t}{1-n}\Big|_0^{\pi}\right]$$

$$\boxed{\begin{aligned}&\therefore\ \sin\ A\cos\ B\\ &= \frac{1}{2}[\sin\ (A+B)\\ &+\sin\ (A-B)]\end{aligned}}$$

$$= \frac{A}{2\pi}\left[\frac{2}{(1+n)} + \frac{2}{1-n}\right] = \frac{2A}{\pi(1-n^2)} \quad \text{for } n \text{ even} \tag{5.55}$$

$$b_n = \frac{2}{T}\int_{t_0}^{t_0+T} x(t)\sin nt\,dt$$

$$= \frac{2}{2\pi}\left[\int_0^{\pi} A \sin t \sin nt\,dt\right] \qquad \boxed{\begin{aligned}&\sin A \sin B\\ &= \frac{1}{2}[\cos(A-B) - \cos(A+B)]\end{aligned}}$$

$$= \frac{A}{2\pi}\left[\int_0^{\pi} \cos(1-n)t - \cos(1+n)t\right]dt$$

$$= \frac{A}{2\pi}\left[\frac{\sin(1-n)t}{1-n}\Big|_0^{\pi} - \frac{\sin(1+n)t}{1+n}\Big|_0^{\pi}\right]$$

$$= 0$$

for $n = 1$

$$a_n = \frac{2}{2\pi}\int_0^{\pi} A \sin t \cos t dt = \frac{A}{2\pi}\left[\int_0^{\pi} \sin 2t dt\right]$$

$$= \frac{-A}{2\pi}\,\frac{\cos 2t}{2}\Big|_0^{\pi} = 0$$

for $n = 1$

$$b_n = \frac{2}{2\pi}\int_0^{\pi} A \sin t \sin t dt = \frac{A}{\pi}\int_0^{\pi} \sin^2 t dt$$

$$= \frac{A}{2\pi}\left[\int_0^{\pi}(1 - \cos 2t)dt\right] = \frac{A}{2}$$

$$A_0 = a_0 = \frac{A}{\pi}$$

$$A_1 = \sqrt{a_1^2 + b_1^2} = A/2$$

$$\boxed{\theta_1 = -\operatorname{Tan}^{-1}\frac{b_1}{a_1} = \frac{-\pi}{2}}$$

$$A_n = \sqrt{a_n^2 + b_n^2} = \frac{2A}{\pi(1 - n^2)}; \quad \theta_n = \tan^{-1}\left(-\frac{b_n}{a_n}\right)$$

$$x(t) = A_0 + \sum_{n=1}^{\infty} A_n \cos(nt + \theta_n)$$

$$x(t) = \frac{A}{\pi} + \frac{A}{2}\cos\left(nt - \frac{\pi}{2}\right) + \sum_{n=2}^{\infty}\frac{2A}{\pi(1 - n^2)}\cos nt$$